

Lab 6: Slow Decay with RC Circuits

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Pre-lab Questions

1. Perform a Kirchhoff analysis of the following RC circuit (Resistor-Capacitor circuit) and determine an equation for the current in the circuit as a function of time, $i(t)$. Calculate the time constant for this circuit.

Using KVL:

$$\mathcal{E} - iR - \frac{Q}{C} = 0.$$

Since $i = dQ/dt$, this becomes a first-order ODE:

$$\mathcal{E} - R \frac{dQ}{dt} - \frac{Q}{C} = 0 \Rightarrow \frac{dQ}{dt} = \frac{C\mathcal{E} - Q}{RC}.$$

Separating variables and integrating (with $Q(0) = 0$):

$$\int_0^Q \frac{dQ'}{C\mathcal{E} - Q'} = \int_0^t \frac{dt'}{RC} \Rightarrow Q(t) = C\mathcal{E}(1 - e^{-t/RC}).$$

Differentiating to get the current:

$$i(t) = \frac{dQ}{dt} = \frac{\mathcal{E}}{R} e^{-t/RC}.$$

The time constant is $\tau = RC$. With $R = 22 \text{ M}\Omega$ and $C = 1 \text{ }\mu\text{F}$:

$$\tau = RC = (22 \cdot 10^6 \text{ }\Omega)(1 \cdot 10^{-6} \text{ F}) = 22 \text{ s}.$$

2. Write a few sentences that explain what the time constant means. Do not use any equations.

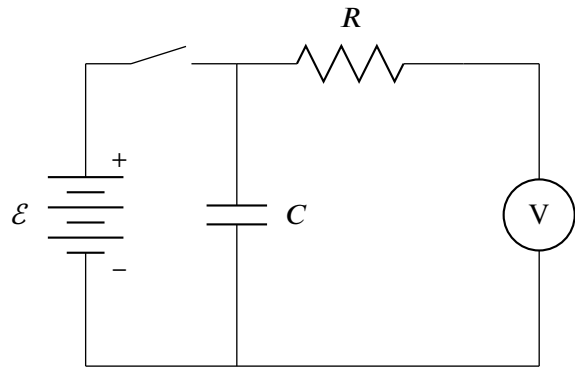
The time constant τ describes how slowly the current in the circuit decays over time. A larger time constant means the current decreases slower, while a smaller time constant means it decays faster.

3. You should already know the definition of the half-life ($\tau_{1/2}$) of a decay. What is the $1/e$ life? Derive the formula for the $1/e$ life. Why is this quantity useful?

The $1/e$ life, $\tau_{1/e}$, is the time it takes for the quantity to fall to $1/e \approx 36.8\%$ of its initial value. For the current $i(t) = i_0 e^{-t/\tau}$, we set $i(\tau_{1/e}) = i_0/e$:

$$i_0 e^{-\tau_{1/e}/\tau} = \frac{i_0}{e} = i_0 e^{-1} \Rightarrow \frac{\tau_{1/e}}{\tau} = 1 \Rightarrow \tau_{1/e} = \tau = RC.$$

The $1/e$ life is useful because it equals the time constant τ directly, so it is more intuitive to calculate and interpret.



Abstract

This experiment investigated the exponential decay of voltage across a capacitor ΔV_C in a discharging RC circuit. A $1\ \mu\text{F}$ capacitor was charged to approximately 45 V using a DC power supply and then discharged through a $22\ \text{M}\Omega$ resistor in series with a DMM (with some internal resistance $\approx 10\ \text{M}\Omega$). The voltage across the DMM was recorded as a function of time, and the data was used to extract the RC time constant τ via two independent methods: measuring the half-life of the decay, and performing a linear regression on a $\ln(V)$ vs. t plot.

From the half-life method, the average half-life (over five measurements) was 24.9 s, yielding $\tau = t_{1/2}/\ln 2 = 35.9$ s. From the logarithmic plot, the slope gave $RC = 38.1$ s. These two values agree to within 5.9%. The DMM internal resistance extracted from $\tau = RC \rightarrow R = \tau/C$ was $15.4\ \text{M}\Omega$, which is within 42.6% of the known value of $10\ \text{M}\Omega$. Sources of uncertainty include human error when starting the stopwatch (the moment at which $t = 0$ may have varied) and the fact that the actual resistance of the DMM is not exactly $10\ \text{M}\Omega$ (the document says it is “about” $10\ \text{M}\Omega$).

Equipment & Procedure

- Component Box
- Digital Multimeter (DMM)
- 45 V DC Power Supply
- Stopwatch¹
- 22 MΩ and 15 MΩ Resistors²
- 1 μF Capacitor

Procedure

1. Direct Measurement of 22 MΩ Resistor

Measured Resistance	$R_{\text{measured}} = 32 \text{ M}\Omega$
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The DMM read 32 MΩ when measuring the 22 MΩ resistor directly. This exceeds the DMM's limit of 20 MΩ, so the measurement isn't considered "accurate".

This also read the DMM's internal resistance, so we can approximate R_{DMM} from this measurement:

$$R_{22 \text{ M}\Omega} + R_{\text{DMM}} = 32 \text{ M}\Omega \rightarrow R_{\text{DMM}} \approx 10 \text{ M}\Omega.$$

This agrees with the document which also states that the DMM internal resistance is about 10 MΩ.

2. Parallel Measurement to Approximate Resistance

We used a resistor rated for 15 MΩ and connected it in parallel to the 22 MΩ resistor:

Measured Resistance of 15 MΩ Resistor	$R_{15 \text{ M}\Omega} = 14.65 \text{ M}\Omega$
Measured Resistance of Equivalent Resistor	$R_{\text{par}} = 8.9 \text{ M}\Omega$
Calculated Resistance	$R_{22 \text{ M}\Omega} = 22.68 \text{ M}\Omega$

Using the parallel resistance formula to solve for $R_{22 \text{ M}\Omega}$:

$$\frac{1}{R_{\text{par}}} = \frac{1}{R_{22 \text{ M}\Omega}} + \frac{1}{R_{15 \text{ M}\Omega}} \Rightarrow R_{22 \text{ M}\Omega} = \frac{R_{\text{par}} R_{15 \text{ M}\Omega}}{R_{15 \text{ M}\Omega} - R_{\text{par}}} = \frac{8.9 \text{ M}\Omega \cdot 14.65 \text{ M}\Omega}{14.65 \text{ M}\Omega - 8.9 \text{ M}\Omega} = 22.68 \text{ M}\Omega.$$

3. Voltage Measurement

Measured Voltage from DMM	13.746 V
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The DMM reads 13.746 V from a 45 V source. The resistor (22.68 MΩ) and DMM (10 MΩ) form a voltage divider, which is why we get about 1/3 of the voltage:

$$V_{\text{DMM}} = \mathcal{E} \cdot \frac{R_{\text{DMM}}}{R_{22 \text{ M}\Omega} + R_{\text{DMM}}} = 45 \text{ V} \cdot \frac{10 \text{ M}\Omega}{22.68 \text{ M}\Omega + 10 \text{ M}\Omega} = 13.77 \text{ V}.$$

¹ A video recording of the DMM with timestamp data was used in place of an actual stopwatch

² For the 15 MΩ Resistor, the measured resistance was 14.65 MΩ

4. Voltage vs. Time Table

A measurement was recorded every time the voltage dropped by approximately half.

V (V)	t (s)
13.746 V	0 s
6.856 V	24.7 s
3.388 V	49.9 s
1.711 V	75.1 s
0.853 V	99.3 s
0.429 V	124.4 s

5. Half-Life Calculations

#	V_{start} (V)	V_{end} (V)	Half-Life (s)
1	13.746 V	6.856 V	24.7 s
2	6.856 V	3.388 V	25.2 s
3	3.388 V	1.711 V	25.2 s
4	1.711 V	0.853 V	24.2 s
5	0.853 V	0.429 V	25.1 s

Average Half-Life	24.9 s
1/e Lifetime ($\tau = t_{1/2} / \ln 2$)	35.9 s
RC (calculated)	$(22.68 \text{ M}\Omega + 10 \text{ M}\Omega) \cdot 1 \text{ }\mu\text{F} = 32.7 \text{ s}$

The half-life is approximately constant across all intervals regardless of the starting voltage, confirming the exponential nature of the decay.

6. DMM Internal Resistance from Half-Life

$$RC = \tau = 35.9 \text{ s} \rightarrow R_{\text{eff}} = \frac{\tau}{C} = \frac{35.9 \text{ s}}{1 \text{ }\mu\text{F}} = 35.9 \text{ M}\Omega.$$

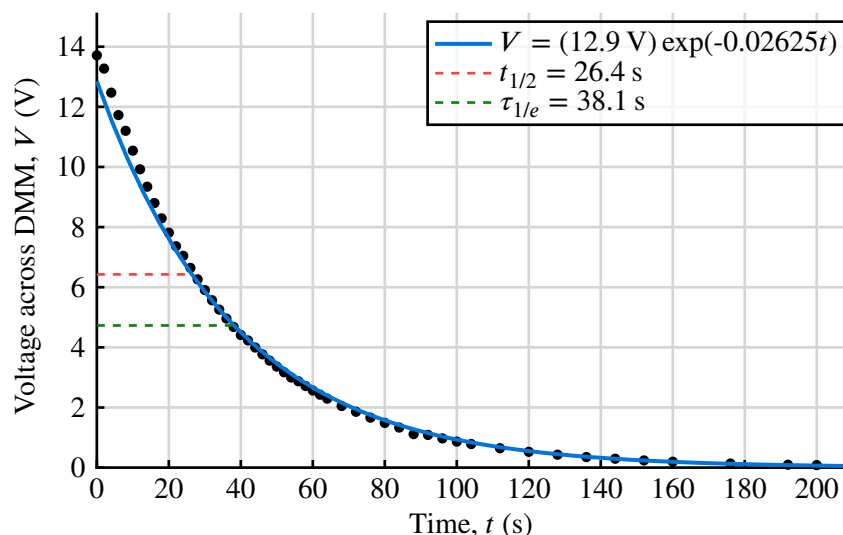
Since $R_{\text{eff}} = R_{22 \text{ M}} + R_{\text{DMM}}$:

$$R_{\text{DMM}} = R_{\text{eff}} - R_{22 \text{ M}} = 35.9 \text{ M}\Omega - 22.68 \text{ M}\Omega = 13.2 \text{ M}\Omega.$$

Calculated DMM Internal Resistance	$R_{\text{DMM}} = 13.2 \text{ M}\Omega$
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Data Analysis

V vs. t (Cartesian Graph)



The graph shows the expected exponential decay shape. The dashed lines indicate the half-life (red, at $V_0/2 \approx 6.43$ V) and the $1/e$ life (green, at $V_0/e \approx 4.73$ V).

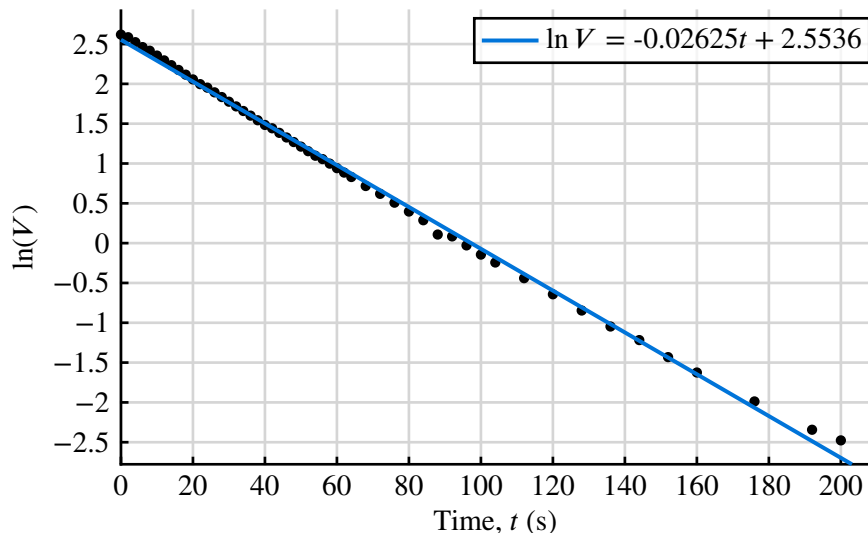
Note: the coefficients in the exponential fit are obtained from the logarithmic graph in the next section.

$\ln(V)$ vs. t (Logarithmic Graph)

Taking the natural logarithm of both sides of $V(t) = V_0 e^{-t/RC}$ gives:

$$\ln V = \ln V_0 - \frac{t}{RC}.$$

This equation tells us that the relationship between $\ln V$ and t is linear with slope $m = -1/RC$ and intercept $b = \ln V_0$.



Performing a linear regression with Desmos, we see that the best fit line for the data has:

$$\text{slope} = m = -0.02625 \text{ s}^{-1}, \quad \text{intercept} = b = 2.5536.$$

The line fits the data very well ($r \approx 0.99$), confirming the exponential decay model.

From the slope:

$$RC = -\frac{1}{m} = -\frac{1}{-0.02625 \text{ s}^{-1}} = 38.1 \text{ s}.$$

From the intercept:

$$V_0 = e^b = e^{2.5536} = 12.9 \text{ V}.$$

Comparison of Methods

Method	RC (s)	Half-Life (s)	R_{DMM} ($\text{M}\Omega$)
Half-Life Method	35.9 s	24.9 s	13.2 $\text{M}\Omega$
Logarithmic Regression	38.1 s	26.4 s	15.4 $\text{M}\Omega$
Calculated (RC)	32.7 s	22.6 s	$\sim 10 \text{ M}\Omega$ (given)

Both experimental methods produce consistent results. The logarithmic slope method is more reliable as it uses more data points over a broader time interval.

Conclusion

Questions

1. Why was the DMM placed in series with the $22\text{ M}\Omega$ resistor instead of in parallel across it?

The DMM has an internal resistance of about $10\text{ M}\Omega$. If placed in parallel with the $22\text{ M}\Omega$ resistor, the combined resistance would be $R_{\text{eq}} = (22\text{ M}\Omega \cdot 10\text{ M}\Omega)/(22\text{ M}\Omega + 10\text{ M}\Omega) \approx 6.9\text{ M}\Omega$, drastically altering the circuit's time constant (since $\tau \propto R$). In series, the DMM measures the voltage across itself (which is proportional to the capacitor voltage through the voltage divider), without significantly changing the total resistance of the discharge path.

2. Start with the defining equations for the Ohm and the Farad and prove that the quantity RC has dimensions of time.

By Ohm's Law $V = IR \rightarrow R = V/I$, we get $1\ \Omega = 1\text{ V/A}$. Also, $1\text{ A} = 1\text{ C/s}$ since $I = dq/dt$. From the definition of capacitance $C = Q/V$, the Farad can be written as $1\text{ F} = 1\text{ C/V}$. Therefore:

$$[RC] = \Omega \cdot \text{F} = \frac{\text{V}}{\text{A}} \cdot \frac{\text{C}}{\text{V}} = \frac{\text{C}}{\text{A}} = \frac{\text{C}}{\text{C/s}} = \text{s}.$$

Thus, RC has dimensions of time.

3. Use your Cartesian graph of V vs. t and write down the mathematical relationship between V (across the DMM) and t and V_0 . Assume that the switch is opened at time $t = 0$ so V_0 is the voltage across the DMM at $t = 0$.

From the Cartesian graph, the voltage decays exponentially:

$$V(t) = V_0 e^{-t/RC},$$

where we found that $V_0 = 12.9\text{ V}$ and $RC = 38.1\text{ s}$ was the best fit our data.

4. Use your mathematical result from question 3 and find a mathematical expression for $I(t)$, the current through the DMM.

By Ohm's Law, the current through the DMM is $I = V/R_{\text{DMM}}$:

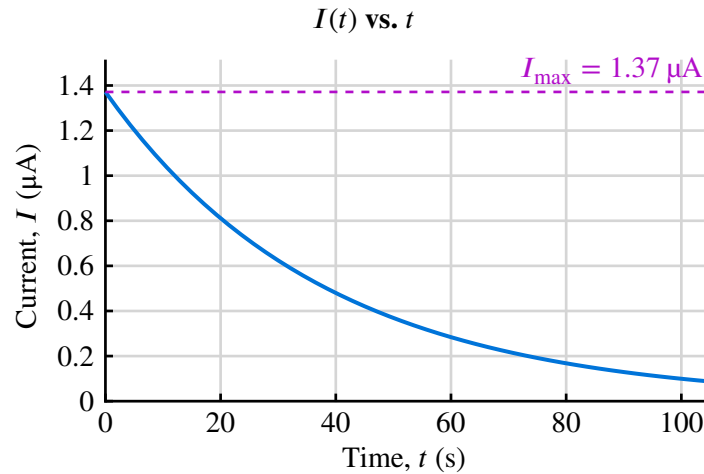
$$I(t) = \frac{V_0}{R_{\text{DMM}}} e^{-t/RC}.$$

This uses the expression we derived in the pre-lab.

5. Assume that the switch opens at $t = 0$. Sketch $I(t)$ vs. t and find a value for the maximum current. Express in microamperes.

The maximum current occurs at $t = 0$:

$$I_{\max} = I(0) = \frac{V_0}{R_{\text{DMM}}} = \frac{12.9 \text{ V}}{10 \text{ M}\Omega} = 1.37 \mu\text{A}.$$



6. Determine the value of the maximum voltage on the capacitor?

When fully charged, no current flows through the circuit. With no current, there is no voltage drop across the resistor or the DMM, so ideally, $V_{\max}$ is just the EMF from the DC power source:

$$V_{\max} = \mathcal{E} = 45 \text{ V}.$$

7. What is the value of the maximum charge stored in the capacitor?

$$Q_{\max} = CV_{\max} = (1 \mu\text{F})(45 \text{ V}) = 45 \mu\text{C}.$$

8. $U = Q^2/2C$. Find the maximum value of the energy stored in the capacitor. Use a ratio to compare this energy to the kinetic energy of a mosquito with a velocity of 0.2 m/s. The mass of a mosquito is about 2.5 mg.

$$U = \frac{Q^2}{2C} = \frac{(4.50 \cdot 10^{-5} \text{ C})^2}{2 \cdot 1.00 \cdot 10^{-6} \text{ F}} = 1.01 \cdot 10^{-3} \text{ J}.$$

The kinetic energy of the mosquito is:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(2.5 \cdot 10^{-6} \text{ kg})(0.2 \text{ m/s})^2 = 5.0 \cdot 10^{-8} \text{ J}.$$

The ratio is:

$$\frac{U}{K} = \frac{1.01 \cdot 10^{-3} \text{ J}}{5.0 \cdot 10^{-8} \text{ J}} = 20,249.$$

The energy stored in the capacitor is roughly $2 \cdot 10^4$ times the kinetic energy of a mosquito.

Error Analysis

Sources of uncertainty include:

1. Human error when starting the stopwatch (the moment at which $t = 0$ may have varied)
 - This would affect all time measurements (half-life, $1/e$ life, and slope) by a constant shift.
2. The actual resistance of the DMM is not exactly $10\text{ M}\Omega$ (the document says it is “about” $10\text{ M}\Omega$).
 - This would affect the voltage divider ratio, which would change the initial voltage V_0 across the DMM and thus affect the current and time constant calculations.